

Basic Differential Amplifier Circuit

Second Formal Report for Understanding Differential Amplifiers & Widlar Current Sources
ECE 333-001 - Electronics 2

Submitted By

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Abstract

This experiment investigated the design and performance of a differential amplifier circuit using a Widlar current source for biasing. The Widlar current source was first verified to ensure proper biasing. The reference current was measured to be around 19.26 mA and the output current to be around 5.55 mA. A differential amplifier was then constructed in conjunction with the current source. The amplifier was verified through measurements to ensure proper biasing conditions. Finally, the full amplifier circuit was tested using both differential and common-mode input conditions. These tests provided measurements for differential gain, common-mode gain, and common-mode rejection ratio (CMRR). The measured differential gain of 73.18 V/V was lower than the expected theoretical value of 260 V/V due to non-ideal conditions. The common-mode gain measured was very small sitting at around 0.00759 and provided a CMRR measurement of roughly 9,641 (79.6 dB). This indicates strong amplification and rejection of noise. Overall, this experiment demonstrates what a practical operation of a differential amplifier should look like.

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INTRODUCTION

The objective of this laboratory was to design, construct, and analyze a single-stage differential amplifier circuit. This circuit uses a previously designed Widlar current source to properly bias the differential amplifier. The first part of this experiment evaluates the current values produced by the previously designed Widlar current source. The second part of the lab focuses on the design and construction of the differential amplifier circuit. This amplifier circuit is designed and constructed “on top” of the Widlar current source, with the current source being used to supply current to the differential amplifier circuit. The final part of the lab uses the full combination of the Widlar current source and the differential amplifier. Chosen input signals are used to evaluate the effectiveness of the complete circuit. This is done by measuring differential (A_d) and common-mode gain (A_{cm}) to achieve a common-mode rejection ratio value (CMRR). This value will help show the effectiveness of the amplifier. The following report covers background/theoretical information, the procedures used during experimentation, and discusses results in relation to what was expected.

BACKGROUND INFORMATION AND THEORY

To better understand what this lab experiment is covering it can be helpful to understand some background information about the circuits used. The two circuits used in this laboratory experiment are a differential amplifier and a Widlar current source. A differential amplifier circuit is a circuit that amplifies the voltage difference between two input voltages, while rejecting “noise” or signals that are common to both inputs. This circuit contains two transistors of the same type (2N2222) and two collector resistors, with a fixed current splitting between the

branches. The collector resistors are used to convert changes in the transistor currents into a measurable output voltage. These are useful for determining the overall gain of the differential amplifier. The amplification happens by a redistribution of current, with the current being distributed based on the value of input voltage applied. So, in theory, if the two input voltages are the same, there should be no differential output since the difference between the two would be zero. When different voltage inputs are applied, the current is redistributed and produces an amplified output based on the differential gain.

A Widlar current source is a modified current mirror. Instead of producing a mirrored current, it produces a much smaller output current compared to the reference current. This brings up the question; what is a current source in the first place? A current source is a circuit that provides a constant current regardless of voltage developed across its terminals. The Widlar current source is made up of two transistors (2N2222) as well as a reference resistor and an emitter resistor. The reference resistor sets a reference current to be compared with the output. The emitter resistor helps produce a smaller output current by introducing a voltage drop to reduce the base-emitter voltage. The combination of all of these components produces a small output current used to bias the differential amplifier circuit.

The most important theory used in this lab is the understanding of differential gain, common-mode gain, and the common-mode rejection ratio. Differential gain is the ratio of output voltage vs. the input voltage difference and represents how much amplification is happening. The smaller the input difference is, the larger the differential gain will be. Common-mode gain is how much the amplifier responds when both the input voltages are equal. The ideal case is zero, but there is typically noise that generates a small gain. The common-mode rejection ratio (CMRR) is a ratio of differential gain vs. common-mode gain. This value indicates

how well the amplifier is rejecting unwanted noise, with a high CMRR value indicating a good rejection of noise and a low CMRR value indicating a less effective rejection of noise.

VERIFICATION OF WIDLAR CURRENT SOURCE

To begin this experiment, the foundation of the design must be tested and verified. The Widlar current source was designed and constructed in a previous lab. The initial design of the Widlar circuit was created with the expectations of a 20 mA reference current while producing roughly 4-6 mA at the output. This portion of the experiment aims to verify that this design still holds up. The circuit schematic is given from the lab manual and is shown below in **Figure 1**.

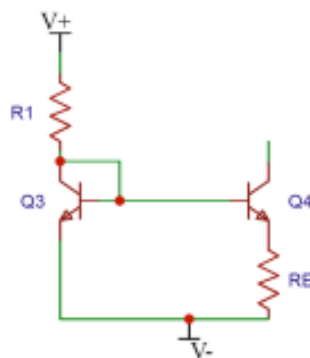


Figure 1. Widlar Current Source Schematic

The layout of the Widlar circuit is clearly shown and the initial design was created to ensure the reference current to be around 20 mA and the output 4-6 mA. The resistors used in this design were the two shown in the schematic as well as a load resistor at the collector of Q4. This load resistor was used to measure the output current. All values of the voltages and resistors were previously designed and can be found below in **Table 1**.

Table 1. Initial Widlar Current Source Values

V+ (V)	V- (V)	R1 (Ω)	RE (Ω)	RL (Ω)
20	-10	1.5k	12.9	1.1k

These design values are used to achieve the desired current values. This first part of the lab aims to verify that the measured currents stay as close to the desired as possible. In order to verify this, measurements are taken over each component. Ohm's law is then used to find and verify the input and output current. The voltages recorded through the collector-emitter terminals of the transistors will help verify if the transistors are acting as expected, which is forward-active mode. The measured values of each component can be found below in **Table 2.**

Table 2. Voltage Readings and Current Values

VR1 (V)	VRL (V)	VRE (V)	VCEQ3 (V)	VCEQ4 (V)	IREF (mA)	IO (mA)
28.9	6.1	0.1	21	0.7	19.26	5.55

The highlighted values were found by using Ohm's law. These current values closely represent the expected values from this design and will allow for a continuation of this experiment. The collector-emitter voltages also prove that both transistors are operating in forward-active mode since they are higher than the saturation voltage of 0.4 volts.

DIFFERENTIAL AMPLIFIER SETUP AND BIASING

After verifying the design of the Widlar current source, the next part of this lab covers the setup and biasing of the differential amplifier circuit. This amplifier circuit is being fed current from the Widlar current source. This differential amplifier is made up of two new 2N2222 transistors and two collector resistors. The full circuit schematic showcasing both the differential amplifier and the Widlar can be found below in **Figure 2**.

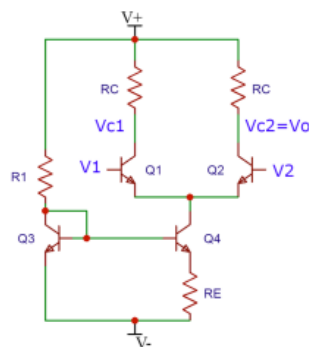


Figure 2. Differential Amplifier Circuit Schematic

The chosen design for the collector resistors was $4.7\text{k}\Omega$. This value was chosen in order to achieve a near 10 volt value at the collector terminals. This design choice was made from the use of Ohm's law coming from the supply rail. Taking into account the 20 volts supplied and the split tail current of 2.775 mA, this leaves a choice of typical resistors around the 3-5k Ω range. The chosen $4.7\text{k}\Omega$ resistors theoretically drop roughly 13 volts over the resistors and leave around 8 at the collector terminal. This design was tested and measured with a multimeter. The results for all measurements are shown below in **Table 3**.

Table 3. Differential Amplifier Voltage Values

VC1=V C2 (V)	VCE1 (V)	VCE2 (V)	VBE1 (V)	VBE2 (V)	VBE4 (V)	VCE4 (V)	RC1 = RC2 (Ω)
8.9	13	12	0.61	0.67	0.65	8.9	4.7k

These results are very close to what was expected, since only common resistor values were used when designing the collector resistors. This was done due to time restraints. With this data collected, it is easy to tell that all the transistors measured are in forward active mode, as their collector-emitter voltages are above saturation levels and the base-emitter is hovering around the turn-on voltage value of roughly 0.6 volts. From this data, the expected differential gain can also be found. Knowing that the collector currents should be half of the tail current provided by the Widlar current source, transconductance can be found and then used to approximate an estimated differential gain. With a collector current of 2.775 and knowing the constant thermal voltage, the transconductance is found to be roughly 0.111 S. Plugging this into the formula for differential gain using $RC = 4.7k\Omega$, the expected differential gain comes out to be **Ad = 260**. All formulas used to solve for this gain are provided in **Appendix A**.

GAIN AND CMRR MEASUREMENTS

For the final part of this lab the whole circuit is no longer being verified, but is instead being used as an actual amplifier in order to determine the differential gain, the common-mode gain, and the common-mode rejection ratio. These things will help determine whether or not this amplifier is properly biased and working as effectively as expected. Since both the prior circuits have been tested and verified near the expected results, the gain should follow closely along with

those expectations. From the last part of the lab, there is an expected differential gain of 260, which will be helpful in the analysis of the recorded results. This final portion of the lab covers two separate experiments. The first experiment applies a differential mode sinusoidal signal across the inputs V1 and V2. The second experiment applies a common-mode sinusoidal signal to both input V1 and V2. The same circuit schematic for the differential amplifier setup is used for this experiment (**Figure 2**).

To apply the differential mode sinusoidal signal, a frequency and amplitude was chosen on the signal generator. The signal was applied by connecting the positive probe to V1 and the negative probe to V2. This ensures a different input at each, giving a difference in the two and allowing for amplification to take place. The frequency and amplitude used is shown below in **Table 4**.

Table 4. Frequency/Amplitude of Applied Signal

Frequency (Hz)	Amplitude (Vpp)
1000	0.02

This frequency and amplitude will be used for both differential and common-mode experiments. After applying this across the two inputs, a signal was produced through an oscilloscope reading both the difference of V1 and V2 as well as the output at VC2. The oscilloscope measurements of the differential mode voltages can be found below in **Figure 3**.

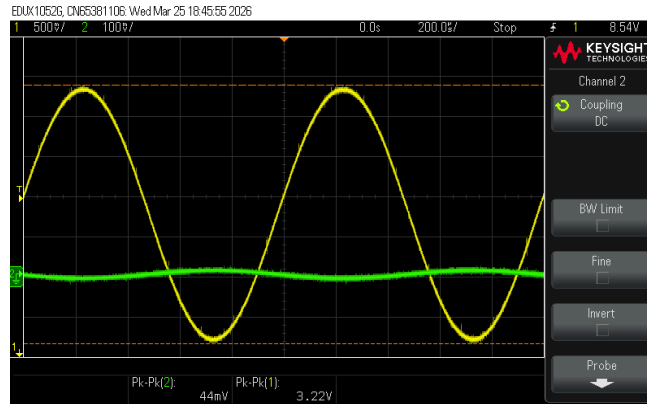


Figure 3. Oscilloscope Reading V_o and V_d

The output shows a large sinusoidal wave oscillating around 10 volts, with the differential voltage being much smaller and oscillating around 0 volts. The measurements and calculated differential gain can be found below in **Table 5**.

Table 5. Differential Mode Voltage Readings

V_o (Vpp)	V_d [$V_2 - V_1$] (Vpp)	A_d (V/V)
3.22	0.044	73.18

After measuring the differential mode voltages, the next experiment requires a similar operation except using a common-mode sinusoidal signal applied to both the inputs. This was accomplished by using two separate signals of the same frequency/amplitude applied at the inputs V_1 and V_2 , then measuring the output voltage at V_{C2} . An oscilloscope was once again used to visualize the common-mode voltages and can be found below in **Figure 4**.

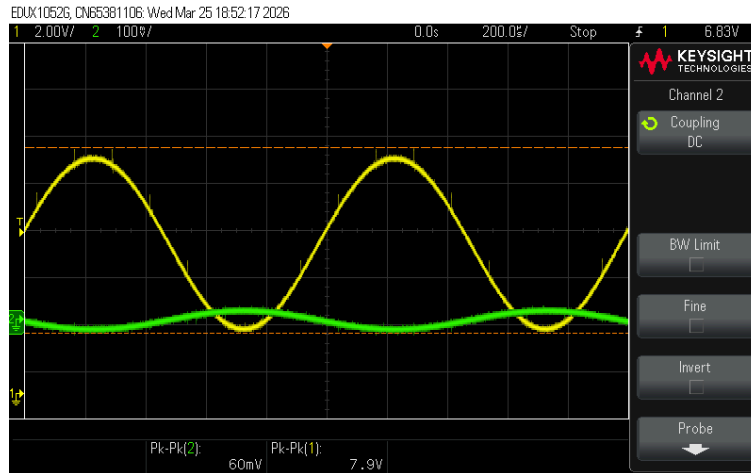


Figure 4. Oscilloscope reading V_o and V_{cm}

The oscilloscope reading of this was similar to the differential mode reading, except the sizes were swapped. The output voltage was very small and oscillated around 10 volts. The common-mode voltage was large in comparison to the output and oscillated around 0 volts. The measurements and calculated common-mode gain can be found below in **Table 6.**

Table 6. Common-Mode Voltage Readings

V_o (Vpp)	V_{cm} (Vpp)	A_{cm} (V/V)
0.060	7.9	0.00759

While the expected differential gain was calculated to be a lot higher than the measured 73.18, calculating the common-mode rejection ratio (CMRR) should still indicate whether or not the amplification is acceptable. Calculating the CMRR from the measured differential and common-mode gain, it calculates to be around **9,641** or **79.6 dB**.

DISCUSSION

The results obtained in this laboratory experiment might not closely align with the expected results found through calculations, but there is a good reason for this. First, the expected differential gain calculated in the second part of this lab was around 260, but the actual differential gain found from measurements was 73.18. This appears to be a big enough difference to consider this experiment to be a failure, but looking at the CMRR achieved from these actual results proves otherwise. While this experiment had many issues with components burning out and breadboards not exactly being cooperative, it is safe to say that these experimental conditions were far from ideal. These expected measurements are under the assumption that everything is ideal and without flaws. This is not how the real world works all the time and is the reason our results vary. Though the results are not as expected, the calculated CMRR shows promising signs of powerful amplification. Though if the differential gain was closer to our expected 260, the CMRR would have been around 90 dB. 79.6 dB is still a very acceptable rejection ratio. There were many things in this lab that could be attributed to the less than accurate amplification: slightly lower collector voltage, slightly lower reference and output current, or the lack of functioning 2N3904 transistors. The results still prove effective amplification and showcase the operations of a functional differential amplifier circuit.

CONCLUSION

This laboratory experiment successfully demonstrates the design, construction, and analysis of a differential amplifier biased by a Widlar current source. The first part of the lab verifies the Widlar current source by measuring voltage values and using Ohm's law to calculate currents. These currents were very close to the expected values, which ensured proper biasing for the differential amplifier circuit. The differential amplifier was then constructed around the

Widlar current source and was designed with the intent of achieving near 10 volts at the collector terminals. It was verified by again measuring voltage values and ensuring forward active operation, confirming proper biasing. The theoretical differential gain was found to be quite a bit higher than the experimental value due to the non-ideal setup/environment. However, the common-mode gain was still small enough to produce an acceptable common-mode rejection ratio of around 80 dB. This CMRR value indicates that the amplifier still effectively amplifies differential signals while rejecting the majority of the noise. Overall, this experiment provided a strong understanding of differential amplifier operation and design.

REFERENCES

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APPENDICES

A. Useful Equations for Differential Gain

$$g_m = \frac{I_C}{V_T} \quad A_d = g_m R_C$$

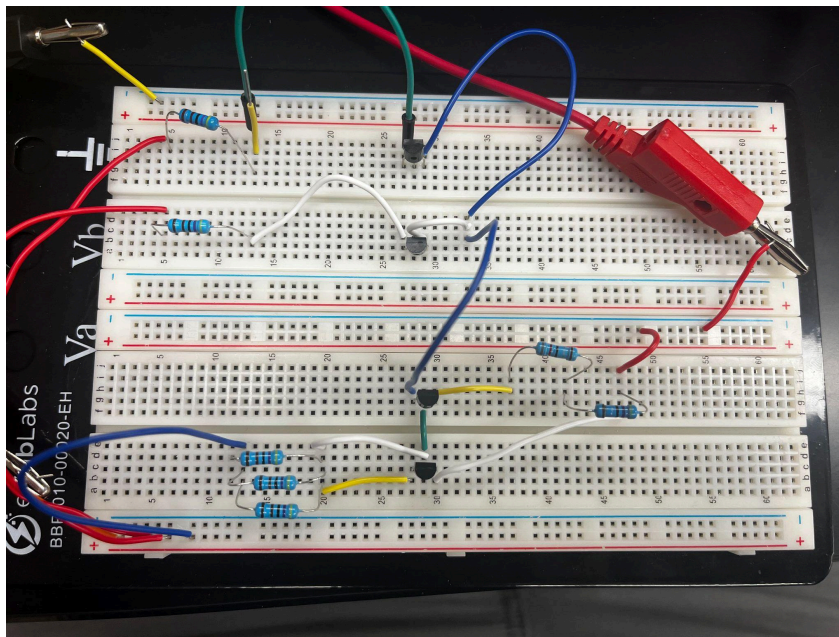
B. Ohm's Law

Ohm's
Law

$$I = \frac{V}{R}$$

Electric current = Voltage / Resistance

C. Breadboard Layout



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